

General Rules:

- You have to create a GitHub repository where to upload your code. Each participant of the group should commit his/her own changes. There should be commit from all the members of the team
- You have to write a 4 (min)- 8(max) pages document in English with the following structure:
 - Introduction to the problem
 - Description of the state of the art (i.e., are there others' works addressing this or a similar problem? What is different this time?)
 - Proposed approach (What procedure have you used? Why?)
 - Results (Plots and tables)
- The document has to be sent at least 3 days before the exam
- You have to create slides for a 5-10 minutes presentation of your work
- The exam will start with a discussion of the project then we will ask general questions on the course subjects.

Project:

Consider this dataset:

https://www.dropbox.com/scl/fi/bm5rgprmkrbmru1fk3cx7/destinationdata_2mesi.tar.gz?rlkey=toeeb09a12up4co2tijss7nq7&dl=0

It records cellular phone movements between different areas in Emilia Romagna (ACE – aree di censimento). The file format is:

layerid;datefrom;dateto;datavalue;toid;toname;Ni;Ns;Tb;Tc;Gm;Gf;F1;F2;F3;F4;F5;F6

in the time interval **datefrom – dateto** there are **datavalue** persons moving from area **layerid** to area **toid**.

Ni;Ns;Tb;Tc;Gm;Gf;F1;F2;F3;F4;F5;F6 are the demographic decomposition of **datavalue**

F1 --> age <18
F2 --> age [18-30]
F3 --> age [31-40]
F4 --> age [41-50]
F5 --> age [51 - 60]
F6 --> age > 60
Gf --> female
Gm --> male
Ns --> not italians
Ni --> italians
Tb --> business
Tc --> consumer

And the file

<https://www.dropbox.com/scl/fi/dt5uyxkzwsjioj3qabac4/layers.csv?rlkey=poysmuj1e2x1nmukrtlym39zh&dl=0>

For each ACE it provides the geometry of the layer in GeoJSON

1. Create visualizations for the data (see folium and geopandas libraires)
2. Create and analyze a forecasting model to predict the movement of users from one area to the other.

Please note that the project description can be rather under-specified. Your task is also to better specify it and discuss your choices in the report.